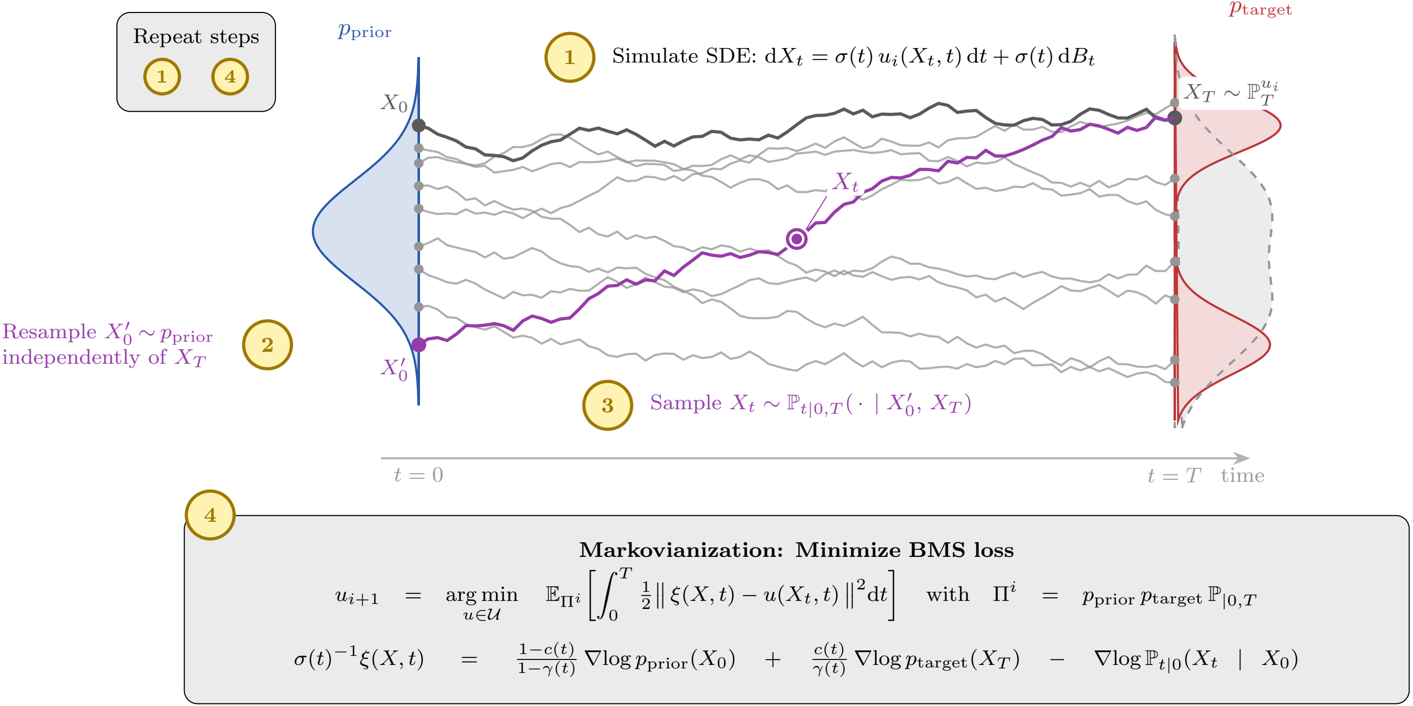

Repeat steps 1 4
p_{prior}
p_{target}
1 Simulate SDE: $\mathrm{d}X_t = \sigma(t)\, u_i(X_t, t)\, \mathrm{d}t + \sigma(t)\, \mathrm{d}B_t$
X_0
$X_T \sim \mathbb{P}_T^{u_i}$
X_t
Resample $X_0' \sim p_{\text{prior}}$ independently of X_T 2
X_0'
3 Sample $X_t \sim \mathbb{P}_{t|0,T}(\,\cdot \mid X_0', X_T)$
$t = 0$
$t = T$ time
4
Markovianization: Minimize BMS loss
$u_{i+1} = \underset{u \in \mathcal{U}}{\arg\min}\ \mathbb{E}_{\Pi^i}\left[\int_0^T \frac{1}{2}\left\|\, \xi(X,t) - u(X_t,t)\, \right\|^2 \mathrm{d}t\right]$ with $\Pi^i = p_{\text{prior}}\, p_{\text{target}}\, \mathbb{P}_{|0,T}$
$\sigma(t)^{-1}\xi(X,t) = \frac{1-c(t)}{1-\gamma(t)}\, \nabla \log p_{\text{prior}}(X_0) + \frac{c(t)}{\gamma(t)}\, \nabla \log p_{\text{target}}(X_T) - \nabla \log \mathbb{P}_{t|0}(X_t \mid X_0)$